STUMPRIDER · DESIGN DECISION RECORD

StumpRider R7 judged on the bamboo local variant

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DOCUMENT CONTROL

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Status	Draft	Date	2026-10-04
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stumprider		
Source	docs/decisions/0004-r7-prototype-mass.md @ 6afbf98		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-04	Amish Chadha	Amish's round-3 decision 2A recorded

0004: R7 judged on the bamboo local variant

- **Date:** 2026-10-04
- **Status:** accepted. Decided by Amish on 2026-10-04: "For round 3, I agree with all your proposed recommendations". For StumpRider this accepts option A of open decision O5 in SMR-DEC-001 (portfolio item 2A).

Context

The aluminium prototype carries 4.28 kg, 0.28 kg over R7's 4 kg (the 4.02 kg quoted earlier was 4.08 kg once lock pins and rivets were counted, and the hook head adds 0.21 kg). The bamboo local variant carries 3.89 kg.

Options considered

A: keep the 32 x 2 aluminium pole and judge R7 on the bamboo local variant. B: a 32 x 1.6 aluminium tube (3.86 kg carried, buckling factor 1.84). C: restate R7 to 4.3 kg. Recommendation was A.

Decision

R7 is judged on the bamboo local variant: 3.89 kg carried, met. The aluminium prototype's 4.28 kg is recorded, not a failure of R7, and is weighed at TRL 4. The pole tube is unchanged.

Consequences

Requirements text, calculation note wording and registers only. No geometry, bill of materials or cost change. Value-engineering target: USD 2,000. Estimated cost of the constructable design: USD 299.50 (USD 1,700.50 under the target).